

Hypothesis Testing

In hypothesis testing data are used to infer which of two competing hypotheses, H_0 or H_1 , is correct. As before, $X \sim P_\theta$ for some $\theta \in \Omega$, and the two competing hypotheses are that the unknown parameter θ lies in set Ω_0 or in set Ω_1 , written

$$H_0 : \theta \in \Omega_0 \text{ versus } H_1 : \theta \in \Omega_1.$$

We assume that Ω_0 and Ω_1 partition Ω , so $\Omega = \Omega_0 \cup \Omega_1$ and $\Omega_0 \cap \Omega_1 = \emptyset$. This chapter derives optimal tests when the parameter θ is univariate. Extensions to higher dimensions are given in Chapter 13.

12.1 Test Functions, Power, and Significance

A *nonrandomized test* of H_0 versus H_1 can be specified by a *critical region* S with the convention that we accept H_1 when $X \in S$ and accept H_0 when $X \notin S$. The performance of this test is described by its *power function* $\beta(\cdot)$, which gives the chance of rejecting H_0 as a function of $\theta \in \Omega$:

$$\beta(\theta) = P_\theta(X \in S).$$

Ideally, we would want $\beta(\theta) = 0$ for $\theta \in \Omega_0$ and $\beta(\theta) = 1$ for $\theta \in \Omega_1$, but in practice this is generally impossible.

In the mathematical formulation for hypothesis testing just presented, the hypotheses H_0 and H_1 have a symmetric role. But in applications H_0 generally represents the status quo, or what someone would believe about θ without compelling evidence to the contrary. In view of this, attention is often focused on tests that have a small chance of error when H_0 is correct. This can be quantified by the *significance level* α defined as

$$\alpha = \sup_{\theta \in \Omega_0} P_\theta(X \in S).$$

In words, the level α is the worst chance of falsely rejecting H_0 .

For technical reasons it is convenient to allow external randomization to “help” the researcher decide between H_0 and H_1 . Randomized tests are characterized by a *test or critical function* φ with range a subset of $[0, 1]$. Given $X = x$, $\varphi(x)$ is the chance of rejecting H_0 . The power function β still gives the chance of rejecting H_0 , and by smoothing,

$$\beta(\theta) = P_\theta(\text{reject } H_0) = E_\theta P_\theta(\text{reject } H_0 | X) = E_\theta \varphi(X).$$

Note that a nonrandomized test with critical region S can be viewed as a randomized test with $\varphi = 1_S$. Conversely, if $\varphi(x)$ is always 0 or 1, then the randomized test with critical function φ can be considered a nonrandomized test with critical region $S = \{x : \varphi(x) = 1\}$.

The set of all critical functions is convex, for if φ_1 and φ_2 are critical functions and $\gamma \in (0, 1)$, then $\gamma\varphi_1 + (1 - \gamma)\varphi_2$ is also a critical function. Convex combinations of nonrandomized tests are not possible, and this is the main advantage of allowing randomization. For randomized tests the level α is defined as

$$\alpha = \sup_{\theta \in \Omega_0} \beta(\theta) = \sup_{\theta \in \Omega_0} E_\theta \varphi(X).$$

12.2 Simple Versus Simple Testing

A hypothesis is called *simple* if it completely specifies the distribution of the data, so $H_i : \theta \in \Omega_i$ is simple when Ω_i contains a single parameter value θ_i . When both hypotheses, H_0 and H_1 are simple, the Neyman–Pearson lemma (Proposition 12.2 below) provides a complete characterization of all reasonable tests. This result makes use of *Lagrange multipliers*, an important idea in optimization theory of independent interest.

Suppose H_0 and H_1 are both simple, and let p_0 and p_1 denote densities for X under H_0 and H_1 , respectively.¹ Since there are only two distributions for the data X , the power function for a test φ has two values,

$$\alpha = E_0 \varphi = \int \varphi(x) p_0(x) d\mu(x)$$

and

$$E_1 \varphi = \int \varphi(x) p_1(x) d\mu(x).$$

Ideally, the first of these values α is near zero, and the other value β is near one. These objectives are in conflict. To do as well as possible we consider the constrained maximization problem of maximizing $E_1 \varphi$ among all test φ with $E_0 \varphi = \alpha$. The following proposition shows that solutions of unconstrained optimization problems with Lagrange multipliers (k) also solve optimization problems with constraints.

¹ As a technical note, there is no loss of generality in assuming densities p_0 and p_1 , since the two distributions P_0 and P_1 are both absolutely continuous with respect to their sum $P_0 + P_1$.

Proposition 12.1. *Suppose $k \geq 0$, φ^* maximizes*

$$E_1\varphi - kE_0\varphi$$

among all critical functions, and $E_0\varphi^ = \alpha$. Then φ^* maximizes $E_1\varphi$ over all φ with level at most α .*

Proof. Suppose φ has level at most α , $E_0\varphi \leq \alpha$. Then

$$\begin{aligned} E_1\varphi &\leq E_1\varphi - kE_0\varphi + k\alpha \\ &\leq E_1\varphi^* - kE_0\varphi^* + k\alpha \\ &= E_1\varphi^*. \end{aligned} \quad \square$$

Maximizing $E_1\varphi - kE_0\varphi$ is fairly easy because

$$\begin{aligned} E_1\varphi - kE_0\varphi &= \int [p_1(x) - kp_0(x)]\varphi(x) d\mu(x) \\ &= \int_{p_1(x) > kp_0(x)} |p_1(x) - kp_0(x)|\varphi(x) d\mu(x) \\ &\quad - \int_{p_1(x) < kp_0(x)} |p_1(x) - kp_0(x)|\varphi(x) d\mu(x). \end{aligned} \quad (12.1)$$

Clearly, any test φ^* maximizing this expression must have

$$\varphi^*(x) = 1, \text{ when } p_1(x) > kp_0(x),$$

and

$$\varphi^*(x) = 0, \text{ when } p_1(x) < kp_0(x).$$

When division by zero is not an issue, these tests are based on the *likelihood ratio* $L(x) = p_1(x)/p_0(x)$, with $\varphi^*(x) = 1$ if $L(x) > k$ and $\varphi^*(x) = 0$ if $L(x) < k$. When $L(x) = k$, $\varphi(x)$ can take any value in $[0, 1]$. Any test of this form is called a *likelihood ratio test*. In addition, the test $\varphi = I\{p_0 = 0\}$ is also considered a likelihood ratio test. (This can be viewed as the test that arises when $k = \infty$.)

Proposition 12.2 (Neyman–Pearson Lemma). *Given any level $\alpha \in [0, 1]$, there exists a likelihood ratio test φ_α with level α , and any likelihood ratio test with level α maximizes $E_1\varphi$ among all tests with level at most α .*

The fact that likelihood ratio tests maximize $E_1\varphi$ among tests with the same or smaller level follows from the discussion above. A formal proof that any desired level $\alpha \in [0, 1]$ can be achieved with a likelihood ratio test is omitted, but similar issues are addressed in the proof of the first part of Theorem 12.9. Also, Example 12.6 below illustrates the type of adjustments that are needed to achieve level α in a typical situation. The next result shows that if a test is optimal, it must be a likelihood ratio test.

Proposition 12.3. Fix $\alpha \in [0, 1]$, let k be the critical value for a likelihood ratio test φ_α described in Proposition 12.2, and define $B = \{x : p_1(x) \neq kp_0(x)\}$. If φ^* maximizes $E_1\varphi$ among all tests with level at most α , then φ^* and φ_α must agree on B , $1_B\varphi^* = 1_B\varphi_\alpha$, a.e. μ .

Proof. Assume $k \in (0, \infty)$ and let $B_1 = \{p_1 > kp_0\}$ and $B_2 = \{p_1 < kp_0\}$, so that $B = B_1 \cup B_2$. Since φ^* and φ_α both maximize $E_1\varphi$, we have $E_1\varphi^* = E_1\varphi_\alpha$. And since φ_α maximizes $E_1\varphi - kE_0\varphi$, $kE_0\varphi_\alpha = k\alpha \leq kE_0\varphi^*$. So $E_0\varphi^*$ must equal α , and φ_α and φ^* both have level α . Thus they both give the same value in (12.1). Since φ_α is 1 on B_1 and 0 on B_2 , using (12.1),

$$\int 1_{B_1}|p_1 - kp_0|(1 - \varphi^*) d\mu + \int 1_{B_2}|p_1 - kp_0|\varphi^* d\mu = 0.$$

Since the arguments of both integrands are nonnegative, both integrands must be zero a.e. μ , and since $|p_1 - kp_0|$ is positive on B_1 and B_2 , we must have

$$1_{B_1}(1 - \varphi^*) + 1_{B_2}\varphi^* = 1_{B_1}|\varphi^* - \varphi_\alpha| + 1_{B_2}|\varphi^* - \varphi_\alpha| = 0$$

a.e. μ .

When $k = 0$, $\varphi_\alpha = 1$ on $p_1 > 0$, and φ_α has power $E_1\varphi_\alpha = 1$. If φ^* has power 1, then $0 = E_1(\varphi_\alpha - \varphi^*) = \int_B |\varphi^* - \varphi_\alpha|p_1 d\mu$, so again φ^* and φ_α agree a.e. μ on B .

For the final degenerate case, “ $k = \infty$,” B should be defined as $\{p_0 > 0\}$. In this case $\varphi_\alpha = 0$ on $p_0 > 0$, and so $\alpha = 0$. If φ^* has level $\alpha = 0$, $0 = E_0(\varphi^* - \varphi_\alpha) = \int_B |\varphi^* - \varphi_\alpha|p_0 d\mu$, and once again φ^* and φ_α agree a.e. μ on B . $\square$

Corollary 12.4. If $P_0 \neq P_1$ and φ_α is a likelihood ratio test with level $\alpha \in (0, 1)$, then $E_1\varphi_\alpha > \alpha$.

Proof. Consider the test φ^* which is identically α , regardless of the value of x . Since φ_α maximizes $E_1\varphi$ among tests with level α , $E_1\varphi_\alpha \geq E_1\varphi^* = \alpha$. Suppose $E_1\varphi_\alpha = \alpha$. Then φ^* also maximizes $E_1\varphi$ among tests with level α , and by Proposition 12.3, φ_α and φ^* must agree a.e. on B . But since $\alpha \in (0, 1)$ and φ_α is 0 or 1 on B , they cannot agree on B . Thus B must be a null set and $p_1 = kp_0$ a.e. μ . Integrating this against μ , k must equal 1, so the densities agree a.e. μ and $P_0 = P_1$. $\square$

Example 12.5. Suppose X is absolutely continuous with density

$$p_\theta(x) = \begin{cases} \theta e^{-\theta x}, & x > 0; \\ 0, & \text{otherwise,} \end{cases}$$

and that we would like to test $H_0 : \theta = 1$ versus $H_1 : \theta = \theta_1$, where θ_1 is a specified constant greater than one. A likelihood ratio test φ is one if

$$\frac{p_1(X)}{p_0(X)} = \frac{\theta_1 e^{-\theta_1 X}}{e^{-X}} > k,$$

or equivalently if

$$X < \frac{\log(\theta_1/k)}{\theta_1 - 1} = k'.$$

The test is zero if $X > k'$. When $X = k'$ the test can take any value in $[0, 1]$, but the choice will not affect any power calculations since $P_\theta(X = k') = 0$. The level of this likelihood ratio test is

$$\alpha = P_0(X < k') = \int_0^{k'} e^{-x} dx = 1 - e^{-k'}.$$

Solving,

$$k' = -\log(1 - \alpha)$$

gives a test with level α . If φ_α is a test with

$$\varphi_\alpha(X) = \begin{cases} 1, & X < -\log(1 - \alpha); \\ 0, & X > -\log(1 - \alpha), \end{cases}$$

then by Proposition 12.1, φ_α maximizes $E_{\theta_1}\varphi$ among all tests with level α . Something surprising and remarkable has happened here. This test φ_α , which is optimal for testing $H_0 : \theta = 1$ versus $H_1 : \theta = \theta_1$, does not depend on the value θ_1 . If φ is any competing test with level α , then

$$E_{\theta_1}\varphi \leq E_{\theta_1}\varphi_\alpha, \quad \text{for all } \theta_1 > 1.$$

Features of this example that give the same optimal test regardless of the value of θ_1 are detailed and exploited in the next section.

Example 12.6. Suppose X has a binomial distribution with success probability θ and $n = 2$ trials. If we are interested in testing $H_0 : \theta = 1/2$ versus $H_1 : \theta = 3/4$, then

$$L(X) = \frac{p_1(X)}{p_0(X)} = \frac{\binom{2}{X}(3/4)^X(1/4)^{2-X}}{\binom{2}{X}(1/2)^X(1/2)^{2-X}} = \frac{3^X}{4}.$$

Under H_0 ,

$$L(X) = \begin{cases} 1/4, & \text{with probability } 1/4; \\ 3/4, & \text{with probability } 1/2; \\ 9/4, & \text{with probability } 1/4. \end{cases}$$

Suppose the desired significance level is $\alpha = 5\%$. If k is less than $9/4$, then $L(2) = 9/4 > k$ and $\varphi(2) = 1$. But then $E_0\varphi(X) \geq \varphi(2)P_0(X = 2) = 1/4$. If

instead k is greater than $9/4$, φ is identically zero. So k must equal $9/4$, and $\varphi(0) = \varphi(1) = 0$. Then to achieve the desired level we must have

$$5\% = E_0\varphi(X) = \frac{1}{4}\varphi(0) + \frac{1}{2}\varphi(1) + \frac{1}{4}\varphi(2) = \frac{1}{4}\varphi(2).$$

Solving, $\varphi(2) = 1/5$ gives a test with level $\alpha = 5\%$.

The assertion in Proposition 12.2 that there exists a likelihood ratio test with any desired level $\alpha \in [0, 1]$ is established in a similar fashion. First k is adjusted so that $P_0(L(X) > k)$ and $P_0(L(X) \geq k)$ bracket α , and then a value $\gamma \in [0, 1]$ is chosen for $\varphi(X)$ when $L(X) = k$ to achieve level α .

12.3 Uniformly Most Powerful Tests

A test φ^* with level α is called *uniformly most powerful* if

$$E_\theta\varphi^* \geq E_\theta\varphi, \quad \forall \theta \in \Omega_1,$$

for all φ with level at most α . Uniformly most powerful tests for composite hypotheses generally only arise when the parameter of interest is univariate, $\theta \in \Omega \subset \mathbb{R}$ and the hypotheses are of the form $H_0 : \theta \leq \theta_0$ versus $H_1 : \theta > \theta_0$, where θ_0 is a fixed constant.² In addition, the family of densities needs to have an appropriate structure.

Definition 12.7. *A family of densities $p_\theta(x)$, $\theta \in \Omega \subset \mathbb{R}$ has monotone likelihood ratios if there exists a statistic $T = T(x)$ such that whenever $\theta_1 < \theta_2$, the likelihood ratio $p_{\theta_2}(x)/p_{\theta_1}(x)$ is a nondecreasing function of T . Also, the distributions should be identifiable, $P_{\theta_1} \neq P_{\theta_2}$ whenever $\theta_1 \neq \theta_2$. Natural conventions concerning division by zero are used here, with the likelihood ratio interpreted as $+\infty$ when $p_{\theta_2} > 0$ and $p_{\theta_1} = 0$. On the null set where both densities are zero the likelihood ratio is not defined and monotonic dependence on T is not required.*

Example 12.8. If the densities p_θ form an exponential family,

$$p_\theta(x) = \exp\{\eta(\theta)T(x) - B(\theta)\}h(x),$$

with $\eta(\cdot)$ strictly increasing, then if $\theta_2 > \theta_1$,

$$\frac{p_{\theta_2}(x)}{p_{\theta_1}(x)} = \exp\{[\eta(\theta_2) - \eta(\theta_1)]T(x) + B(\theta_1) - B(\theta_2)\},$$

which is increasing in $T(x)$.

² Minor variants are possible here: H_0 could be $\theta = \theta_0$, $\theta < \theta_0$, $\theta \geq \theta_0$, etc. Uniformly most powerful tests are also possible when the null hypothesis H_0 is two-sided, but this case sees little application.

Theorem 12.9. *Suppose the family of densities has monotone likelihood ratios. Then*

1. *The test φ^* given by*

$$\varphi^*(x) = \begin{cases} 1, & T(x) > c; \\ \gamma, & T(x) = c; \\ 0, & T(x) < c, \end{cases}$$

is uniformly most powerful testing $H_0 : \theta \leq \theta_0$ versus $H_1 : \theta > \theta_0$ and has level $\alpha = E_{\theta_0}\varphi^$. Also, the constants $c \in \mathbb{R}$ and $\gamma \in [0, 1]$ can be adjusted to achieve any desired significance level $\alpha \in (0, 1)$.*

2. *The power function $\beta(\theta) = E_{\theta}\varphi^*$ for this test is nondecreasing and strictly increasing whenever $\beta(\theta) \in (0, 1)$.*
3. *If $\theta_1 < \theta_0$, then this test φ^* minimizes $E_{\theta_1}\varphi$ among all tests with $E_{\theta_0}\varphi = \alpha = E_{\theta_0}\varphi^*$.*

Proof. Suppose $\theta_1 < \theta_2$ and let

$$L(x) = \frac{p_{\theta_2}(x)}{p_{\theta_1}(x)}.$$

Since the family has monotone likelihood ratios, L is a nondecreasing function of T . If k is the value of L when $T = c$, then (see [Figure 12.1](#))

$$\varphi^*(x) = \begin{cases} 1, & \text{when } L > k; \\ 0, & \text{when } L < k. \end{cases}$$

Thus φ^* is a likelihood ratio test of $\theta = \theta_1$ versus $\theta = \theta_2$. By Corollary 12.4, $E_{\theta_2}\varphi^* \geq E_{\theta_1}\varphi^*$, with strict inequality unless both expectations are zero or one. So the second assertion of the theorem holds, and φ^* has level $\alpha = E_{\theta_0}\varphi^*$.

To show that φ^* is uniformly most powerful, suppose $\tilde{\varphi}$ has level at most α and $\theta_1 > \theta_0$. Then $E_{\theta_0}\tilde{\varphi} \leq \alpha$, and since φ^* is a likelihood ratio test of $\theta = \theta_0$ versus $\theta = \theta_1$ maximizing $E_{\theta_1}\varphi$ among all tests with $E_{\theta_0}\varphi \leq E_{\theta_0}\varphi^* = \alpha$, $E_{\theta_1}\varphi^* \geq E_{\theta_1}\tilde{\varphi}$. Similarly, if $\theta_1 < \theta_0$, since φ^* is a likelihood ratio test of $\theta = \theta_1$ versus $\theta = \theta_0$ with some critical value k , it must maximize $E_{\theta_0}\varphi - kE_{\theta_1}\varphi$. Thus if $\tilde{\varphi}$ is a competing test with $E_{\theta_0}\tilde{\varphi} = \alpha = E_{\theta_0}\varphi^*$, then $E_{\theta_1}\tilde{\varphi} \geq E_{\theta_1}\varphi^*$, proving the third assertion in the theorem.

To finish, we must show that c and γ can be adjusted so that $E_{\theta_0}\varphi^* = \alpha$. Let F denote the cumulative distribution function for T when $\theta = \theta_0$. Define

$$c = \sup\{x : F(x) \leq 1 - \alpha\}.$$

If $x > c$, then $F(x) > 1 - \alpha$. Because F is right continuous,

$$F(c) = \lim_{x \downarrow c} F(x) \geq 1 - \alpha.$$

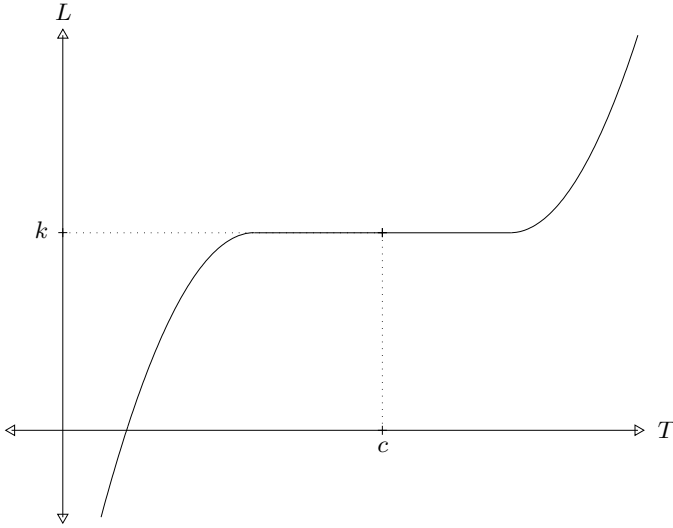


Fig. 12.1. The likelihood ratio L as a function of T .

But for $x < c$, $F(x) \leq 1 - \alpha$, and so

$$F(c-) \stackrel{\text{def}}{=} \lim_{x \uparrow c} F(x) \leq 1 - \alpha.$$

Now let $q = F(c) - F(c-) = P_{\theta_0}(T = c)$ (see Problem 1. 16), and define

$$\gamma = \frac{F(c) - (1 - \alpha)}{q}.$$

(If $q = 0$, γ can be any value in $[0, 1]$.) By the bounds for $F(c)$ and $F(c-)$, γ must lie in $[0, 1]$, and then

$$\begin{aligned} E_{\theta_0} \varphi^* &= \gamma P_{\theta_0}(T = c) + P_{\theta_0}(T > c) \\ &= F(c) - (1 - \alpha) + (1 - F(c)) \\ &= \alpha. \end{aligned}$$

□

Example 12.10. Suppose our data $X_1, \dots, X_n$ are i.i.d. from the uniform distribution on $(0, \theta)$. The joint density $p_\theta(x)$ is positive if and only if $x_i \in (0, \theta)$, $i = 1, \dots, n$, and this happens if and only if $M(x) = \min\{x_1, \dots, x_n\} > 0$ and $T(x) = \max\{x_1, \dots, x_n\} < \theta$. Thus

$$p_\theta(x) = \begin{cases} 1/\theta^n, & M(x) > 0 \text{ and } T(x) < \theta; \\ 0, & \text{otherwise.} \end{cases}$$

Suppose $\theta_2 > \theta_1$, $M(x) > 0$, and $T(x) < \theta_2$. Then

$$\frac{p_{\theta_2}(x)}{p_{\theta_1}(x)} = \begin{cases} \theta_1^n / \theta_2^n, & T(x) < \theta_1; \\ +\infty, & T(x) \geq \theta_1. \end{cases}$$

This shows that the family of joint densities has monotone likelihood ratios. (The behavior of the likelihood ratio when both densities are zero does not matter; this is why there is no harm assuming $M(x) > 0$ and $T(x) < \theta_2$.) If we are interested in testing $H_0 : \theta \leq 1$ versus $H_1 : \theta > 1$, the test function φ given by

$$\varphi = \begin{cases} 1, & T \geq c; \\ 0, & \text{otherwise.} \end{cases}$$

is uniformly most powerful. This test has level

$$P_1(T \geq c) = 1 - c^n,$$

and a specified level α can be achieved taking

$$c = (1 - \alpha)^{1/n}.$$

The power of this test is

$$\beta_\varphi(\theta) = P_\theta(T \geq c) = \begin{cases} 0, & \theta < c; \\ 1 - \frac{1 - \alpha}{\theta^n}, & \theta \geq c. \end{cases}$$

In this example, one competing test $\tilde{\varphi}$ is given by

$$\tilde{\varphi} = \begin{cases} \alpha, & T < 1; \\ 1, & T \geq 1. \end{cases}$$

For $\theta < 1$, $E_\theta \tilde{\varphi} = \alpha$, so this test also has level α . For $\theta > 1$, this test has power

$$\begin{aligned} \beta_{\tilde{\varphi}}(\theta) &= E_\theta \tilde{\varphi} = \alpha P_\theta(T < 1) + P_\theta(T \geq 1) \\ &= \frac{\alpha}{\theta^n} + 1 - \frac{1}{\theta^n} \\ &= \beta_\varphi(\theta). \end{aligned}$$

The power functions β_φ and $\beta_{\tilde{\varphi}}$ are plotted in [Figure 12.2](#). Because the power functions for $\tilde{\varphi}$ and φ are the same under H_1 , these two tests are both uniformly most powerful. Under H_0 , the power function for φ is smaller than the power function for $\tilde{\varphi}$, so φ is certainly the better test. The test $\tilde{\varphi}$ here is an example of an *inadmissible*³ uniformly most powerful test.

³ A test $\tilde{\varphi}$ is called inadmissible if a competing test φ has a better power function: $\beta_{\tilde{\varphi}}(\theta) \geq \beta_\varphi(\theta)$ for all $\theta \in \Omega_0$, and $\beta_{\tilde{\varphi}}(\theta) \leq \beta_\varphi(\theta)$ for all $\theta \in \Omega_1$, with strict inequality in one of these bounds for some $\theta \in \Omega$.

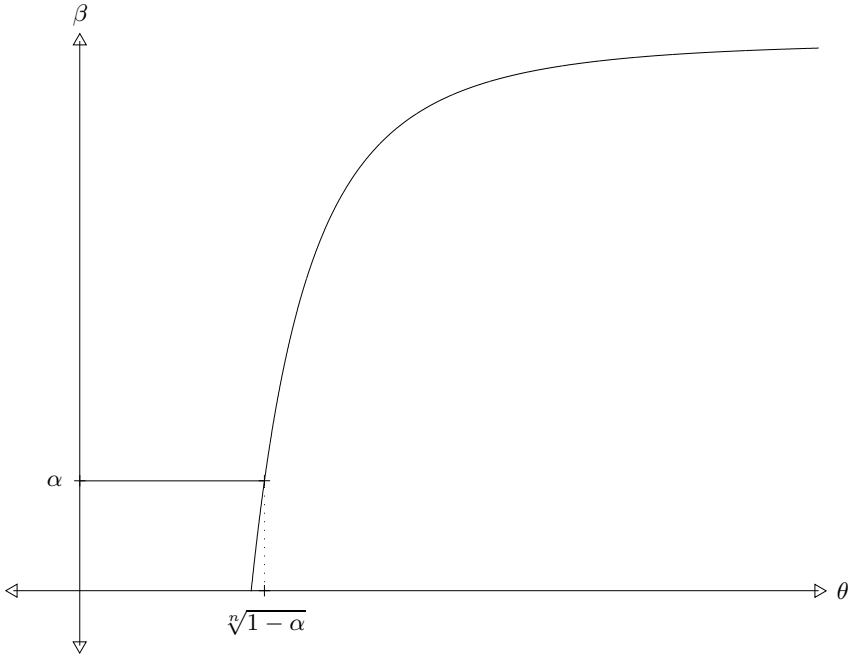


Fig. 12.2. Power functions β_φ and $\beta_{\tilde{\varphi}}$.

12.4 Duality Between Testing and Interval Estimation

Recall that a random set $S(X)$ is a $1 - \alpha$ confidence region for a parameter $\xi = \xi(\theta)$ if

$$P_\theta(\xi \in S(X)) \geq 1 - \alpha, \quad \forall \theta \in \Omega.$$

For every ξ_0 , let $A(\xi_0)$ be the acceptance region for a nonrandomized level α test of $H_0 : \xi(\theta) = \xi_0$ versus $H_1 : \xi(\theta) \neq \xi_0$, so that

$$P_\theta[X \in A(\xi(\theta))] \geq 1 - \alpha, \quad \forall \theta \in \Omega.$$

Define

$$S(x) = \{\xi : x \in A(\xi)\}.$$

Then $\xi(\theta) \in S(X)$ if and only if $X \in A(\xi(\theta))$, and so

$$P_\theta(\xi(\theta) \in S(X)) = P_\theta(X \in A(\xi(\theta))) \geq 1 - \alpha.$$

This shows that $S(X)$ is a $1 - \alpha$ confidence region for ξ .

The construction above can be used to construct confidence regions from a family of nonrandomized tests. Conversely, a $1 - \alpha$ confidence region $S(X)$ can be used to construct a family of tests. If we define

$$\varphi = \begin{cases} 1, & \xi_0 \notin S(X); \\ 0, & \text{otherwise,} \end{cases}$$

then if $\xi(\theta) = \xi_0$,

$$E_\theta \varphi = P_\theta(\xi_0 \notin S(X)) = P_\theta(\xi(\theta) \notin S(X)) \leq \alpha.$$

This shows that this test has level at most α testing $H_0 : \xi(\theta) = \xi_0$ versus $H_1 : \xi(\theta) \neq \xi_0$. If the coverage probability for $S(X)$ is exactly $1 - \alpha$, $P_\theta(\xi(\theta) \in S(X)) = 1 - \alpha$, for all $\theta \in \Omega$, then φ will have level exactly α .

Example 12.11. Suppose the densities for a model have monotone likelihood ratios. Also, for convenience assume $F_\theta(t) = P_\theta(T \leq t)$ is continuous and strictly increasing in t , for all $\theta \in \Omega$. For each $\theta \in \Omega$, define $u(\theta)$ so that

$$P_\theta(T < u(\theta)) = F_\theta(u(\theta)) = 1 - \alpha.$$

Then

$$\varphi = \begin{cases} 1, & T \geq u(\theta_0); \\ 0, & \text{otherwise,} \end{cases}$$

is uniformly most powerful testing $H_0 : \theta = \theta_0$ versus $H_1 : \theta > \theta_0$ and has level

$$E_{\theta_0} \varphi = P_{\theta_0}(T \geq u(\theta_0)) = \alpha.$$

This test has acceptance region

$$A(\theta_0) = \{x : T(x) < u(\theta_0)\}.$$

Proposition 12.12. *The function $u(\cdot)$ is strictly increasing.*

Proof. Suppose $\theta > \theta_0$. By the second part of Theorem 12.9, the power function for φ is strictly increasing at θ_0 , and so

$$E_\theta \varphi = P_\theta(T \geq u(\theta_0)) > E_{\theta_0} \varphi = \alpha.$$

Thus $P_\theta(T < u(\theta_0)) < 1 - \alpha$. But from the definition of $u(\cdot)$, $P_\theta(T < u(\theta)) = 1 - \alpha$, and so $u(\theta) > u(\theta_0)$. Since $\theta > \theta_0$ are arbitrary parameter values, u is strictly increasing. $\square$

The confidence set dual to the family of tests with acceptance regions $A(\theta)$, $\theta \in \Omega$, is

$$S(X) = \{\theta : X \in A(\theta)\} = \{\theta : T(X) < u(\theta)\}.$$

Because u is strictly increasing, this region is the interval (see [Figure 12.3](#))

$$S(X) = (u^\leftarrow(T), \infty) \cap \Omega.$$

Here $u^\leftarrow$ is the inverse function of u .

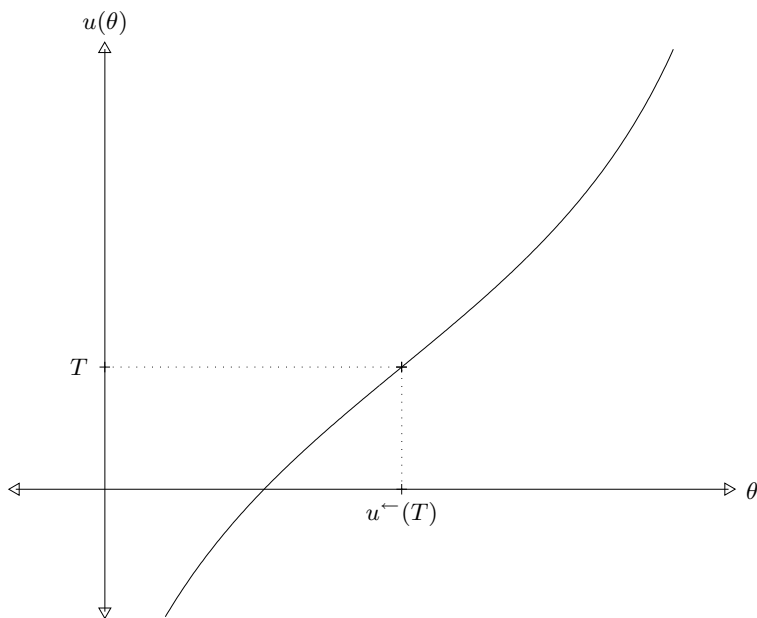


Fig. 12.3. The increasing function u .

For a concrete example, suppose X is exponential with mean θ , so

$$p_{\theta}(x) = \frac{1}{\theta} e^{-x/\theta}, \quad x > 0.$$

The densities for X form an exponential family with $\eta = -1/\theta$, an increasing function of θ . So we have monotone likelihood ratios with $T = X$. The function u is defined so that

$$P_{\theta}(X < u(\theta)) = 1 - \alpha$$

or

$$P_{\theta}(X \geq u(\theta)) = \alpha.$$

Because $P_{\theta}(X \geq x) = e^{-x/\theta}$,

$$e^{-u(\theta)/\theta} = \alpha.$$

Solving,

$$u(\theta) = -\theta \log \alpha.$$

Since $u^{-}(X)$ is the value θ solving

$$X = -\theta \log \alpha,$$

we have

$$u^{\leftarrow}(X) = \frac{X}{-\log \alpha},$$

and the $1 - \alpha$ confidence set for θ is

$$S(X) = \left(\frac{X}{-\log \alpha}, \infty \right).$$

As a check,

$$\begin{aligned} P_{\theta}(\theta \in S(X)) &= P_{\theta} \left(\frac{X}{-\log \alpha} < \theta \right) \\ &= \int_0^{-\theta \log \alpha} \frac{1}{\theta} e^{-x/\theta} dx \\ &= 1 - \alpha. \end{aligned}$$

Our construction of confidence sets by duality works with any family of level α tests. But intuition suggests that better tests should give better confidence intervals. In the example just considered, tests in the family are uniformly most powerful, and a natural conjecture would be that the dual confidence interval $S(X)$ should be optimal in some related sense. This is indeed the case. To deduce the proper notion of optimality, let $S^*(X)$ be a competing confidence set, and take

$$\varphi^* = \begin{cases} 1, & \theta_0 \notin S^*(X); \\ 0, & \text{otherwise.} \end{cases}$$

Then φ^* is a test of $H_0 : \theta = \theta_0$ with level at most α . The corresponding test φ dual to $S(X)$ is

$$\varphi = \begin{cases} 1, & \theta_0 \notin S(X); \\ 0, & \text{otherwise.} \end{cases}$$

If the tests dual to S are uniformly most powerful testing $H_0 : \theta = \theta_0$ versus $H_1 : \theta > \theta_0$, then for any $\theta > \theta_0$,

$$E_{\theta} \varphi \geq E_{\theta} \varphi^*.$$

The left- and right-hand sides of this equation are $P_{\theta}(\theta_0 \notin S(X))$ and $P_{\theta}(\theta_0 \notin S^*(X))$, respectively, and so

$$P_{\theta}(\theta_0 \in S(X)) \leq P_{\theta}(\theta_0 \in S^*(X)). \quad (12.2)$$

This shows that if θ is the true value of the parameter, then $S(X)$ has a smaller chance of containing any incorrect value $\theta_0 < \theta$.

In practice, a researcher may be most concerned with the length of a confidence interval, and the optimality for S in (12.2) may seem less relevant.

However, using Fubini's theorem, there is a relation between the expected length and the probabilities $P_\theta(\theta_0 \in S(X))$. Let λ denote Lebesgue measure on $\mathbb{R}$, so that $\lambda(A) = \int_A dx$ is the length of A . Also, assume for convenience that Ω is the interval $(\underline{\omega}, \overline{\omega})$ (we allow $\underline{\omega} = -\infty$ and/or $\overline{\omega} = \infty$). Then, by Fubini's theorem,

$$\begin{aligned} E_\theta \lambda(S(X) \cap (\underline{\omega}, \theta)) &= E_\theta \int_{\underline{\omega}}^{\theta} I(\theta_0 \in S(X)) d\theta_0 \\ &= \int \int_{\underline{\omega}}^{\theta} I(\theta_0 \in S(x)) d\theta_0 dP_\theta(x) \\ &= \int_{\underline{\omega}}^{\theta} P_\theta(\theta_0 \in S(X)) d\theta_0. \end{aligned}$$

Similarly,

$$E_\theta \lambda(S^*(X) \cap (\underline{\omega}, \theta)) = \int_{\underline{\omega}}^{\theta} P_\theta(\theta_0 \in S(X)) d\theta_0,$$

and so, by (12.2),

$$E_\theta \lambda(S(X) \cap (\underline{\omega}, \theta)) \leq E_\theta \lambda(S^*(X) \cap (\underline{\omega}, \theta)).$$

So the expected length of $S(X)$ below θ is minimal among all $1 - \alpha$ confidence intervals.

12.5 Generalized Neyman–Pearson Lemma⁴

Treatment of two-sided hypotheses in the next section relies on an extension of the Neyman–Pearson lemma in which the test function must satisfy several constraints. Let $g(x)$ take values in $\mathbb{R}^m$, and consider maximizing

$$\int \varphi f d\mu$$

over all test functions φ satisfying

$$\int \varphi g d\mu = c, \tag{12.3}$$

where c is a specified vector in $\mathbb{R}^m$. Introducing a Lagrange multiplier $k \in \mathbb{R}^m$, consider maximizing

$$\int (f - k \cdot g) \varphi d\mu \tag{12.4}$$

⁴ Results in the rest of this chapter and Chapter 13 are more technical and are not used in subsequent chapters.

without constraint. A test function maximizing (12.4) will have form

$$\varphi(x) = \begin{cases} 1, & f(x) > k \cdot g(x); \\ 0, & f(x) < k \cdot g(x), \end{cases} \quad (12.5)$$

for a.e. x (μ). As in our discussion of the Neyman–Pearson lemma, if a function of this form satisfies (12.3), it clearly solves the constrained maximization problem.

Let $\mathcal{K}$ denote the set of all test functions (measurable functions with range a subset of $[0, 1]$).

Theorem 12.13. *Assume f and g are both integrable with respect to μ and that the class $\mathcal{C}$ of all test functions $\varphi \in \mathcal{K}$ satisfying (12.3) is not empty. Then*

1. *There exists a test function φ^* maximizing $\int \varphi f d\mu$ over $\mathcal{C}$.*
2. *If $\varphi^* \in \mathcal{C}$ satisfies (12.5) for some $k \in \mathbb{R}^m$, then φ^* maximizes $\int \varphi f d\mu$ over $\mathcal{C}$.*
3. *If $\varphi^* \in \mathcal{C}$ has form (12.5) with $k \geq 0$, then φ^* maximizes $\int \varphi f d\mu$ over all φ satisfying $\int \varphi g d\mu \leq c$.*
4. *Let $\mathcal{L}_g$ be the linear mapping from test functions $\varphi \in \mathcal{K}$ to vectors in $\mathbb{R}^m$ given by $\mathcal{L}_g(\varphi) = \int \varphi g d\mu$, and let M denote the range of $\mathcal{L}_g$. Then M is closed and convex. If c lies in the interior of M , there exists a Lagrange multiplier $k \in \mathbb{R}^m$ and a test function $\varphi^* \in \mathcal{C}$ maximizing $\int (f - k \cdot g) \varphi d\mu$ over $\varphi \in \mathcal{K}$. Also, if any $\varphi \in \mathcal{C}$ maximizes $\int \varphi f d\mu$ over $\mathcal{C}$, then (12.5) must hold a.e. μ .*

The proof of this result relies on an important and useful result from functional analysis, the weak compactness theorem. In functional analysis, functions are viewed as points in a vector space, much as vectors are viewed as points in $\mathbb{R}^n$. But notions of convergence for functions are much richer. For instance, functions f_n , $n \geq 1$, converge pointwise to f if $\lim_{n \rightarrow \infty} f_n(x) = f(x)$ for all x . In contrast, uniform convergence would hold if $\lim_{n \rightarrow \infty} \sup_x |f_n(x) - f(x)| = 0$. Uniform convergence implies pointwise convergence, but not vice versa. (For instance, the functions $1_{(n, n+1)}$ converge pointwise to the zero function, but the convergence is not uniform.) The notion of convergence of interest here is called *weak convergence*.

Definition 12.14. *A sequence of uniformly bounded measurable functions φ_n , $n \geq 1$, converge weakly to φ , written $\varphi_n \xrightarrow{w} \varphi$, if*

$$\int \varphi_n f d\mu \rightarrow \int \varphi f d\mu$$

whenever $\int |f| d\mu < \infty$.

If the functions φ_n converge pointwise to φ , then weak convergence follows from dominated convergence, but pointwise convergence is not necessary for weak convergence. With this notion of convergence, the objective function in Theorem 12.13,

$$\mathcal{L}_f(\varphi) = \int \varphi f d\mu,$$

is a continuous function of φ ; that is, $\mathcal{L}_f(\varphi_n) \rightarrow \mathcal{L}_f(\varphi)$ whenever $\varphi_n \xrightarrow{w} \varphi$. The linear constraint function $\mathcal{L}_g$ introduced in Theorem 12.13, is also continuous.

Theorem 12.15 (Weak Compactness Theorem). *The set $\mathcal{K}$ is weakly compact.⁵ any sequence of functions φ_n , $n \geq 1$, in $\mathcal{K}$ has a convergent subsequence, $\varphi_{n(j)} \xrightarrow{w} \varphi \in \mathcal{K}$ as $j \rightarrow \infty$.*

See Appendix A.5 of Lehmann and Romano (2005) for a proof. In the proof of Theorem 12.13, we also need the following result, called the *supporting hyperplane theorem*. For this and other results in convex analysis, see Rockafellar (1970).

Theorem 12.16 (Supporting Hyperplane Theorem). *If x is a point on the boundary of a convex set $K \subset \mathbb{R}^m$, then there exists a nonzero vector $v \in \mathbb{R}^m$ such that*

$$v \cdot y \leq v \cdot x, \quad \forall y \in K.$$

Proof of Theorem 12.13. The first assertion follows by weak compactness of $\mathcal{K}$. Take

$$K_{\mathcal{C}} = \sup_{\varphi \in \mathcal{C}} \mathcal{L}_f(\varphi),$$

and let φ_n , $n \geq 1$, be a sequence of test functions in $\mathcal{C}$ such that

$$\mathcal{L}_f(\varphi_n) \rightarrow K_{\mathcal{C}}.$$

By the weak compactness theorem (Theorem 12.15), there must be a subsequence $\varphi_{n(m)}$, $m \geq 1$, with

$$\varphi_{n(m)} \xrightarrow{w} \varphi^* \in \mathcal{K},$$

and since $\mathcal{L}_f$ is continuous, $\mathcal{L}_f(\varphi^*) = K_{\mathcal{C}}$. If $\varphi^* \in \mathcal{C}$ we are done. But this follows by continuity of $\mathcal{L}_g$ because $\mathcal{L}_g(\varphi^*) = \lim_{m \rightarrow \infty} \mathcal{L}_g(\varphi_{n(m)})$ and $\mathcal{L}_g(\varphi_{n(m)}) = c$ for all $m \geq 1$.

For the second assertion, if $\varphi^* \in \mathcal{C}$ has form (12.5), then φ^* maximizes $\int \varphi(f - k \cdot g) d\mu = \mathcal{L}_f(\varphi) - k \cdot \mathcal{L}_g(\varphi)$ over all $\mathcal{K}$, and hence φ^* maximizes $\mathcal{L}_f(\varphi) - k \cdot \mathcal{L}_g(\varphi)$ over $\varphi \in \mathcal{C}$. But when $\varphi \in \mathcal{C}$, $\mathcal{L}_f(\varphi) - k \cdot \mathcal{L}_g(\varphi) = \mathcal{L}_f(\varphi) - k \cdot c$, and so φ^* maximizes $\mathcal{L}_f(\varphi)$ over $\varphi \in \mathcal{C}$.

⁵ The topology of weak convergence has a countable base, and so compactness and sequential compactness (stated in this theorem) are equivalent.

Suppose $\varphi^* \in \mathcal{C}$ has form (12.5), so it maximizes $\mathcal{L}_f(\varphi) - k \cdot \mathcal{L}_g(\varphi)$ over all $\varphi \in \mathcal{K}$. Then if $k \geq 0$ and $\mathcal{L}_g(\varphi) \leq c$,

$$\begin{aligned}\mathcal{L}_f(\varphi) &\leq \mathcal{L}_f(\varphi) - k \cdot \mathcal{L}_g(\varphi) + k \cdot c \\ &\leq \mathcal{L}_f(\varphi^*) - k \cdot \mathcal{L}_g(\varphi^*) + k \cdot c = \mathcal{L}_f(\varphi^*).\end{aligned}$$

This proves the third assertion.

The final assertion is a bit more involved. First, M is convex, for if $x_0 = \mathcal{L}_g(\varphi_0)$ and $x_1 = \mathcal{L}_g(\varphi_1)$ are arbitrary points in M , and if $\gamma \in [0, 1]$, using the linearity of $\mathcal{L}_g$,

$$\begin{aligned}\gamma x_0 + (1 - \gamma)x_1 &= \gamma \mathcal{L}_g(\varphi_0) + (1 - \gamma)\mathcal{L}_g(\varphi_1) \\ &= \mathcal{L}_g(\gamma\varphi_0 + (1 - \gamma)\varphi_1) \in M.\end{aligned}$$

Closure of M follows by weak compactness and continuity of $\mathcal{L}_g$. Suppose x is a limit point of M , so that $x = \lim_{n \rightarrow \infty} \mathcal{L}_g(\varphi_n)$ for some sequence of test functions φ_n , $n \geq 1$. Letting $\varphi_{n(m)}$, $m \geq 1$, be a subsequence converging weakly to φ ,

$$\mathcal{L}_g(\varphi) = \lim_{m \rightarrow \infty} \mathcal{L}_g(\varphi_{n(m)}) = \lim_{m \rightarrow \infty} x_{n(m)} = x,$$

which shows that $x \in M$.

For the final part of the theorem, assume that c lies in the interior of M . Let $\varphi^* \in \mathcal{C}$ maximize $\mathcal{L}_f(\varphi)$ over $\varphi \in \mathcal{C}$, and take $K_{\mathcal{C}} = \mathcal{L}_f(\varphi^*)$. Define $\mathcal{L} : \mathcal{K} \rightarrow \mathbb{R}^{m+1}$ by

$$\mathcal{L}(\varphi) = \begin{pmatrix} \mathcal{L}_f(\varphi) \\ \mathcal{L}_g(\varphi) \end{pmatrix}.$$

The arguments showing that M is closed and convex also show that the range $\tilde{M}$ of $\mathcal{L}$ is closed and convex. The point

$$x = \mathcal{L}(\varphi^*) = \begin{pmatrix} \mathcal{L}_f(\varphi^*) \\ \mathcal{L}_g(\varphi^*) \end{pmatrix} = \begin{pmatrix} \mathcal{L}_f(\varphi^*) \\ c \end{pmatrix}$$

lies in $\tilde{M}$. Because φ^* maximizes $\mathcal{L}_f(\varphi)$ over $\varphi \in \mathcal{C}$, if $\epsilon > 0$, the point

$$\begin{pmatrix} \mathcal{L}_f(\varphi^*) + \epsilon \\ c \end{pmatrix}$$

cannot lie in $\tilde{M}$, and thus x lies on the boundary of $\tilde{M}$. By the supporting hyperplane theorem (Theorem 12.16), there is a nonzero vector $v = \begin{pmatrix} a \\ b \end{pmatrix}$ such that

$$v \cdot y \leq v \cdot x, \quad \forall y \in \tilde{M},$$

or, equivalently, such that

$$a\mathcal{L}_f(\varphi) + b \cdot \mathcal{L}_g(\varphi) \leq a\mathcal{L}_f(\varphi^*) + b \cdot \mathcal{L}_g(\varphi^*), \quad \forall \varphi \in \mathcal{K}.$$

Here a cannot be zero, for then this bound would assert that $b \cdot \mathcal{L}_g(\varphi) \leq b \cdot c$ for all $\varphi \in \mathcal{K}$, contradicting the assumption that c lies in the interior of M . For $\varphi \in \mathcal{C}$ this bound becomes $a\mathcal{L}_f(\varphi) \leq a\mathcal{L}_f(\varphi^*)$. Because φ^* maximizes $\mathcal{L}_f(\varphi)$ over $\varphi \in \mathcal{C}$, a must be positive, unless we are in a degenerate situation in which $\mathcal{L}_f(\varphi) = \mathcal{L}_f(\varphi^*)$ for all $\varphi \in \mathcal{C}$. And if a is positive, we are done, for then the bound is

$$\mathcal{L}_f(\varphi) + (b/a) \cdot \mathcal{L}_g(\varphi) \leq \mathcal{L}_f(\varphi^*) + (b/a) \cdot \mathcal{L}_g(\varphi^*), \quad \forall \varphi \in \mathcal{K},$$

and we can take $k = -b/a$.

To handle the degenerate case, suppose $\mathcal{L}_g(\varphi_1) = \mathcal{L}_g(\varphi_2) = \tilde{c} \neq c$. Because c is an interior point of M , it can be expressed as a nontrivial convex combination of $\tilde{c}$ and some other point $\mathcal{L}_g(\varphi_3) \neq c$ in M ; that is,

$$c = \gamma\tilde{c} + (1 - \gamma)\mathcal{L}_g(\varphi_3),$$

for some $\gamma \in (0, 1)$. Since $\mathcal{L}_g$ is linear,

$$\gamma\varphi_1 + (1 - \gamma)\varphi_3 \quad \text{and} \quad \gamma\varphi_2 + (1 - \gamma)\varphi_3$$

both lie in $\mathcal{C}$, and so

$$\begin{aligned} \mathcal{L}_f(\gamma\varphi_1 + (1 - \gamma)\varphi_3) &= \gamma\mathcal{L}_f(\varphi_1) + (1 - \gamma)\mathcal{L}_f(\varphi_3) \\ &= \mathcal{L}_f(\gamma\varphi_2 + (1 - \gamma)\varphi_3) \\ &= \gamma\mathcal{L}_f(\varphi_2) + (1 - \gamma)\mathcal{L}_f(\varphi_3). \end{aligned}$$

So we must have $\mathcal{L}_f(\varphi_1) = \mathcal{L}_f(\varphi_2)$. Thus, if $(\frac{\ell_0}{\tilde{c}})$ and $(\frac{\ell_1}{\tilde{c}})$ both lie in $\tilde{M}$, then $\ell_0 = \ell_1$. Since $\tilde{M}$ is convex and contains the origin, the only way this can happen is if $\mathcal{L}_f(\varphi)$ is a linear function of $\mathcal{L}_g(\varphi)$,

$$\mathcal{L}_f(\varphi) = k \cdot \mathcal{L}_g(\varphi), \quad \varphi \in \mathcal{K}.$$

In this case, φ^* trivially maximizes $\mathcal{L}_f(\varphi) - k \cdot \mathcal{L}_g(\varphi)$.

To finish, if φ maximizes $\mathcal{L}_f$ over $\mathcal{C}$, then $\mathcal{L}(\varphi) = \mathcal{L}(\varphi^*)$ and φ also maximizes $\mathcal{L}_f - k \cdot \mathcal{L}_g$ over $\mathcal{K}$. It is then clear that (12.5) must hold a.e. μ ; if not, a function satisfying (12.5) would give a larger value for $\mathcal{L}_f - k \cdot \mathcal{L}_g$. $\square$

12.6 Two-Sided Hypotheses

This section focuses on testing $H_0 : \theta = \theta_0$ versus $H_1 : \theta \neq \theta_0$ with data from a one-parameter exponential family. Generalization to families satisfying a condition analogous to the monotone likelihood ratio condition is possible. Tests of $H_0 : \theta \in [\theta_1, \theta_2]$ versus $H_1 : \theta < \theta_1$ or $\theta > \theta_2$ can be developed in a similar fashion, but results about the point null hypothesis seem more useful in practice. Also, uniformly most powerful tests when H_0 is two-sided can be obtained (see Problem 12.39), but these tests are not used very often.

With data from an exponential family there will be a sufficient statistic, and the next result shows that we can then restrict attention to tests based on the sufficient statistic.

Theorem 12.17. *Suppose that T is sufficient for the model $\mathcal{P} = \{P_\theta : \theta \in \Omega\}$. Then for any test $\varphi = \varphi(X)$, the test*

$$\psi = \psi(T) = E_\theta[\varphi(X) \mid T]$$

has the same power function as φ ,

$$E_\theta \psi(T) = E_\theta \varphi(X), \quad \forall \theta \in \Omega.$$

Proof. This follows immediately from smoothing,

$$E_\theta \varphi(X) = E_\theta E_\theta[\varphi(X) \mid T] = E_\theta \psi(T). \quad \square$$

The next theorem shows that if the densities for X come from an exponential family, then the densities for T will also be from an exponential family. This is established using the following fundamental lemma, which shows how likelihood ratios can be introduced to write an expectation under one distribution as an expectation under a different distribution. This lemma is quite useful in a variety of situations.

Lemma 12.18. *Let P_0 and P_1 be possible distributions for a random vector X with densities p_0 and p_1 with respect to μ . If $p_1(x) = 0$ whenever $p_0(x) = 0$, then $P_1 \ll P_0$ and P_1 has density*

$$\frac{dP_1}{dP_0}(x) = L(x) = \frac{p_1(x)}{p_0(x)}$$

with respect to P_0 . (The value for $L(x)$ when $p_0(x) = 0$ does not matter; for definiteness, take $L(x) = 1$ when $p_0(x) = 0$.) Introducing this likelihood ratio, we can write expectations under P_1 as expectations under P_0 using the formula

$$E_1 h(X) = E_0 h(X) L(X),$$

valid whenever the expectations exist. When h is an indicator function, we have

$$P_1(B) = E_0 1_B(X) L(X).$$

Proof. First note that $N = \{x : p_0(x) = 0\}$ is a null set for P_0 because

$$P_0(N) = \int_N p_0 d\mu = \int 1_N p_0 d\mu = \int 0 d\mu = 0.$$

Similarly, $\{x : p_1(x) = 0\}$ is a null set for P_1 , and since $N \subset \{x : p_1(x) = 0\}$, N is also a null set for P_1 . So $1_{N^c} = 1$ a.e. P_0 and P_1 , and multiplication by

this function cannot change the value of an integral against either distribution. Suppose M is a null set for P_0 . Then

$$\int 1_M p_0 d\mu = \int 1_M 1_{N^c} p_0 d\mu = \int 1_{M \cap N^c} p_0 d\mu,$$

which implies that $1_{M \cap N^c} p_0 = 0$ a.e. μ (by the second fact about integration in Section 1.4). Because $p_0 > 0$ whenever the indicator is 1, $M \cap N^c$ must be a null set for μ . But P_1 is dominated by μ , and so $M \cap N^c$ is a null set for P_1 . But $M \subset N \cup (M \cap N^c)$, which is a union of two null sets for P_1 , showing that M must be a null set for P_1 . To write expectations under P_1 as expectations under P_0 ,

$$\begin{aligned} E_1 h(X) &= \int h p_1 d\mu = \int 1_{N^c} h p_1 d\mu = \int 1_{N^c} h \frac{p_1}{p_0} p_0 d\mu \\ &= \int 1_{N^c} h L p_0 d\mu = \int h L p_0 d\mu = E_0 h(X) L(X). \quad \square \end{aligned}$$

Theorem 12.19. *If the distribution for X comes from an exponential family with densities*

$$p_\theta(x) = h(x) e^{\eta(\theta) \cdot T(x) - B(\theta)}, \quad \theta \in \Omega,$$

then the induced distribution for $T = T(X)$ has density

$$q_\theta(t) = e^{\eta(\theta) \cdot t - B(\theta)}, \quad \theta \in \Omega,$$

with respect to some measure ν .

Proof. Two ideas are used. First, using Lemma 12.18, we can introduce a likelihood ratio to write probabilities under P_θ as expectations under P_{θ_0} , where θ_0 is a fixed point in Ω . This likelihood ratio is

$$L = \frac{p_\theta(X)}{p_{\theta_0}(X)} = e^{[\eta(\theta) - \eta(\theta_0)] \cdot T + B(\theta_0) - B(\theta)}.$$

The second is that expectations of functions of T can be written as integrals against the density of X , or as integrals against the marginal distribution of T . Let ν^* denote the marginal distribution of T when $\theta = \theta_0$. Then

$$\begin{aligned} P_\theta(T \in B) &= E_{\theta_0} I\{T \in B\} e^{[\eta(\theta) - \eta(\theta_0)] \cdot T + B(\theta_0) - B(\theta)} \\ &= \int I\{t \in B\} e^{[\eta(\theta) - \eta(\theta_0)] \cdot t + B(\theta_0) - B(\theta)} d\nu^*(t) \\ &= \int_B q_\theta(t) e^{-\eta(\theta_0) \cdot t + B(\theta_0)} d\nu^*(t). \end{aligned}$$

If we define ν by

$$\nu(A) = \int_A e^{-\eta(\theta_0) \cdot t + B(\theta_0)} d\nu^*(t),$$

then ν has density $e^{-\eta(\theta_0) \cdot t + B(\theta_0)}$ with respect to ν^* , and

$$P_\theta(T \in B) = \int_B q_\theta(t) d\nu(t),$$

completing the proof. $\square$

Consider testing $H_0 : \theta = \theta_0$ versus $H_1 : \theta \neq \theta_0$ based on data X with density

$$h(x)e^{\eta(\theta)T(x) - B(\theta)}, \quad \theta \in \Omega, \quad (12.6)$$

where η is strictly increasing and differentiable. From results in Section 12.3, there are level α tests $\varphi_\pm$ with form

$$\varphi_+ = \begin{cases} 1, & T > c_+; \\ \gamma_+, & T = c_+; \\ 0, & T < c_+, \end{cases} \quad \text{and} \quad \varphi_- = \begin{cases} 1, & T < c_-; \\ \gamma_-, & T = c_-; \\ 0, & T > c_-. \end{cases}$$

These tests are most powerful for one-sided alternatives. If $\theta_- < \theta_0 < \theta_+$, then φ_+ will have maximal power at θ_+ , and φ_- will have maximal power at θ_- . Since these tests are different, this shows that there cannot be a uniformly most powerful level α test. To achieve uniformity, we must restrict the class of tests under consideration. We do this by constraining the derivative of the power function at θ_0 . The formula in the following theorem is useful.

Theorem 12.20. *If η is differentiable at θ and θ lies in the interior of Ω , then the derivative of the power function β for a test φ is given by*

$$\beta'(\theta) = \eta'(\theta)E_\theta T\varphi - B'(\theta)\beta(\theta).$$

Proof. If differentiation under the integral sign works, then

$$\begin{aligned} \beta'(\theta) &= \frac{\partial}{\partial \theta} \int \varphi(x) e^{\eta(\theta)T(x) - B(\theta)} h(x) d\mu(x) \\ &= \int \varphi(x) \frac{\partial}{\partial \theta} e^{\eta(\theta)T(x) - B(\theta)} h(x) d\mu(x) \\ &= \int \varphi(x) [\eta'(\theta)T(x) - B'(\theta)] e^{\eta(\theta)T(x) - B(\theta)} h(x) d\mu(x) \\ &= \eta'(\theta)E_\theta \varphi T - B'(\theta)E_\theta \varphi, \end{aligned}$$

and the result follows. Differentiation under the integral sign can be justified using Theorem 2.4 and the chain rule, or by dominated convergence directly. $\square$

Note that because φ_+ has maximal power for $\theta > \theta_0$,

$$\begin{aligned}\beta'_\varphi(\theta_0) &= \lim_{\epsilon \downarrow 0} \frac{\beta_\varphi(\theta_0 + \epsilon) - \beta_\varphi(\theta_0)}{\epsilon} \\ &\leq \lim_{\epsilon \downarrow 0} \frac{\beta_{\varphi_+}(\theta_0 + \epsilon) - \beta_{\varphi_+}(\theta_0)}{\epsilon} = \beta'_{\varphi_+}(\theta_0) \stackrel{\text{def}}{=} m_+.\end{aligned}\quad (12.7)$$

Similarly, $\beta'_\varphi(\theta_0) \geq \beta'_{\varphi_-}(\theta_0) \stackrel{\text{def}}{=} m_-$. For $m \in [m_-, m_+]$, let $\mathcal{C}_m$ denote the class of all level α tests φ with $\beta'_\varphi(\theta_0) = m$. The next theorem shows that when $m \in (m_-, m_+)$ there is a uniformly most powerful test in $\mathcal{C}_m$. This test is two-sided, according to the following definition.

Definition 12.21. A test φ is called two-sided if there are finite constants $t_1 \leq t_2$ such that

$$\varphi = \begin{cases} 1, & T < t_1 \text{ or } T > t_2; \\ 0, & T \in (t_1, t_2). \end{cases}$$

In addition, the test should not be one-sided. Specifically, $E\varphi I\{T \geq t_2\}$ and $E\varphi I\{T \leq t_1\}$ should both be positive.

Theorem 12.22. If θ_0 lies in the interior of Ω , $\alpha \in (0, 1)$, X has density (12.6), and η is differentiable and strictly increasing with $0 < \eta'(\theta_0) < \infty$, then for any value $m \in (m_-, m_+)$ there is a two-sided test $\varphi^* \in \mathcal{C}_m$. Any such test is uniformly most powerful in class $\mathcal{C}_m$: for any competing test $\varphi \in \mathcal{C}_m$,

$$E_\theta \varphi \leq E_\theta \varphi^*, \quad \forall \theta \in \Omega.$$

Proof. Using Theorem 12.20, if $\varphi \in \mathcal{C}_m$, then

$$\beta'_\varphi(\theta_0) = \eta'(\theta_0) E_{\theta_0} T \varphi - \alpha B'(\theta_0) = m,$$

which happens if and only if

$$E_{\theta_0} T \varphi = \int \varphi T p_{\theta_0} d\mu = \frac{m + \alpha B'(\theta_0)}{\eta'(\theta_0)}.$$

If we define

$$g(x) = \left(\frac{p_{\theta_0}(x)}{T(x)p_{\theta_0}(x)} \right) \quad \text{and} \quad c = \left(\frac{\alpha}{m + \alpha B'(\theta_0)} \right),$$

then a test function φ lies in $\mathcal{C}_m$ if and only if $\mathcal{L}_g(\varphi) = \int \varphi g d\mu = c$. Because $m_+ > m$ and $m_- < m$, the point c lies in the interior of the convex hull of the four points $\mathcal{L}_g(\varphi_+)$, $\mathcal{L}_g(\varphi_-)$, $\mathcal{L}_g(\alpha - \epsilon)$, and $\mathcal{L}_g(\alpha + \epsilon)$. (Here “ $\alpha \pm \epsilon$ ” denotes a test function that equals $\alpha \pm \epsilon$ regardless of the value of X .) Thus c lies in the interior of the range M of $\mathcal{L}_g$.

With this background on the nature of the constraints for test functions φ in $\mathcal{C}_m$, we can now use the last assertion in Theorem 12.13 to show that there exists a two-sided test φ^* in $\mathcal{C}_m$. Let $\tilde{\theta}$ be some fixed point in Ω different from θ_0 and consider maximizing

$$E_{\tilde{\theta}}\varphi = \int \varphi p_{\tilde{\theta}} d\mu$$

over φ in $\mathcal{C}_m$. Using the fourth assertion in Theorem 12.13 there is a Lagrange multiplier $k \in \mathbb{R}^2$ and a test $\varphi^* \in \mathcal{C}_m$ maximizing $\int \varphi(p_{\tilde{\theta}} - k \cdot g) d\mu$ with form

$$\varphi^* = \begin{cases} 1, & p_{\tilde{\theta}} > (k_1 + k_2 T)p_{\theta_0}; \\ 0, & p_{\tilde{\theta}} < (k_1 + k_2 T)p_{\theta_0}. \end{cases}$$

Dividing through by p_{θ_0} ,

$$\varphi^* = \begin{cases} 1, & \exp\{(\eta(\tilde{\theta}) - \eta(\theta_0))T - B(\tilde{\theta}) + B(\theta_0)\} > k_1 + k_2 T; \\ 0, & \exp\{(\eta(\tilde{\theta}) - \eta(\theta_0))T - B(\tilde{\theta}) + B(\theta_0)\} < k_1 + k_2 T. \end{cases}$$

The line $k_1 + k_2 t$ must intersect the exponential function $\exp\{(\eta(\tilde{\theta}) - \eta(\theta_0))t - B(\tilde{\theta}) + B(\theta_0)\}$, for otherwise φ^* would be identically one. Because the exponential function is strictly convex, the line and exponential function intersect either once, if the line is tangent to the curve, or twice. Let $t_1^* < t_2^*$ denote the two points of intersection when the line is not tangent, and let $t_1^* = t_2^*$ be the single point of intersection when the line is tangent to the exponential function. Since the exponential function is convex, φ^* has form

$$\varphi^* = \begin{cases} 1, & T < t_1^* \text{ or } T > t_2^*; \\ 0, & T \in (t_1^*, t_2^*). \end{cases}$$

To finish showing that φ^* is a two-sided test, we need to verify φ^* is not one-sided. Suppose $E_{\theta}\varphi I\{T \leq t_1\} = 0$. By Theorem 12.9 $\psi = E_{\theta}(\varphi^*|T)$ has the same power function as φ^* , and this test is uniformly most powerful testing $\theta \leq \theta_0$ against $\theta > \theta_0$ by Theorem 12.9. Because φ_+ is also uniformly most powerful, the power functions for φ^* and φ_+ must agree for $\theta \geq \theta_0$, and the slope of the power function for φ^* at θ_0 must be m_+ . This is a contradiction for if φ^* lies in $\mathcal{C}_m$ this slope must be $m < m_+$. Similarly, $E_{\theta}\varphi I\{T \geq t_2\} = 0$, and so φ^* is a two-sided test.

To conclude, we show that any two-sided test $\tilde{\varphi} \in \mathcal{C}_m$ is uniformly most powerful. So assume

$$\tilde{\varphi} = \begin{cases} 1, & T < \tilde{t}_1 \text{ or } T > \tilde{t}_2; \\ 0, & T \in (\tilde{t}_1, \tilde{t}_2), \end{cases}$$

and let θ be an arbitrary point in Ω not equal to θ_0 . Define $\kappa \in \mathbb{R}^2$ so that the line $\kappa_1 + \kappa_2 t$ passes through the points

$$\left(\tilde{t}_1, e^{[\eta(\theta) - \eta(\theta_0)]\tilde{t}_1 - B(\theta) + B(\theta_0)} \right)$$

and

$$\left(\tilde{t}_2, e^{[\eta(\theta) - \eta(\theta_0)]\tilde{t}_2 - B(\theta) + B(\theta_0)} \right).$$

If $\tilde{t}_1 = \tilde{t}_2$, so these points are the same, then the line should also have slope $(\theta - \theta_0)e^{(\theta - \theta_0)\tilde{t}_1 - A(\theta) + A(\theta_0)}$ so that it lies tangent to the exponential curve. By convexity of the exponential function and algebra similar to that used above, $\tilde{\varphi}$ has form

$$\tilde{\varphi} = \begin{cases} 1, & p_\theta > (\kappa_1 + \kappa_2 T)p_{\theta_0}; \\ 0, & p_\theta < (\kappa_1 + \kappa_2 T)p_{\theta_0}. \end{cases}$$

From this, $\tilde{\varphi}$ clearly maximizes $\int \varphi(p_\theta - \kappa \cdot g) d\mu$ over all $\varphi \in \mathcal{K}$. But for test function $\varphi \in \mathcal{C}_m$,

$$\int \varphi(p_\theta - \kappa \cdot g) d\mu = E_\theta \varphi - \kappa \cdot c.$$

Thus $E_\theta \tilde{\varphi} \geq E_\theta \varphi$ for any $\varphi \in \mathcal{C}_m$, and, since θ is arbitrary, $\tilde{\varphi}$ is uniformly most powerful in $\mathcal{C}_m$. $\square$

Remark 12.23. A similar result can be obtained testing $H_0 : \theta \in [\theta_1, \theta_2]$ versus $H_1 : \theta \notin [\theta_1, \theta_2]$. Suppose φ^* is a two-sided test with $E_{\theta_1} \varphi^* = \alpha_1$ and $E_{\theta_2} \varphi^* = \alpha_2$. Then φ^* has level $\alpha = \max\{\alpha_1, \alpha_2\}$ and is uniformly most powerful among all tests φ with $E_{\theta_1} \varphi = \alpha_1$ and $E_{\theta_2} \varphi = \alpha_2$.

Remark 12.24. If the slope m for the power function for a test φ at θ_0 differs from zero, then there will be points $\theta \neq \theta_0$ where the power for the test is less than α . If this happens, the test φ is called *biased*. If an unbiased test is desired, the slope m should be constrained to equal zero. This idea is developed and extended in the next section of this chapter.

12.7 Unbiased Tests

In the previous section we encountered a situation in which uniformly most powerful tests cannot exist unless we constrain the class of test functions under consideration. One appealing constraint restricts attention to tests that are unbiased according to the following definition. Theorem 12.26 below finds uniformly most powerful unbiased tests for one-parameter exponential families, and Chapter 13 has extensions to higher dimensions.

Definition 12.25. A test φ for $H_0 : \theta \in \Omega_0$ versus $H_1 : \theta \in \Omega_1$ with level α is unbiased if its power $\beta_\varphi(\theta) = E_\theta \varphi$ satisfies

$$\beta_\varphi(\theta) \leq \alpha, \quad \forall \theta \in \Omega_0 \quad \text{and} \quad \beta_\varphi(\theta) \geq \alpha, \quad \forall \theta \in \Omega_1.$$

If there is a uniformly most powerful test φ^* , then it is automatically unbiased because $\beta_{\varphi^*}(\theta) \geq \beta_{\varphi}(\theta)$, for all $\theta \in \Omega_1$, and the right-hand side of this inequality is identically α for the degenerate test, which equals α regardless of the observed data.

Theorem 12.26. *If $\alpha \in (0, 1)$, θ_0 lies in the interior of Ω , X has density (12.6), and η is differentiable and strictly increasing with $0 < \eta'(\theta_0) < \infty$, then there is a two-sided, level α test φ^* with $\beta'_{\varphi^*}(\theta_0) = 0$. Any such test is uniformly most powerful testing $H_0 : \theta = \theta_0$ versus $H_1 : \theta \neq \theta_0$ among all unbiased tests with level α .*

Changing the sign of T and η , this result is also true if η is differentiable and strictly decreasing with $-\infty < \eta'(\theta_0) < 0$.

Proof. Since θ_0 lies in the interior of Ω , the power function for any unbiased test φ must have zero slope at θ_0 , and so $\varphi \in \mathcal{C}_0$. The theorem is essentially a corollary of Theorem 12.22, provided $0 \in (m_-, m_+)$. This is established in the following lemma. $\square$

Lemma 12.27. *Under the assumptions of Theorem 12.26,*

$$m_+ = \beta'_{\varphi_+}(\theta_0) > 0 \text{ and } m_- = \beta'_{\varphi_-}(\theta_0) < 0.$$

Proof. Let us begin showing that $E_{\theta_0}T\varphi_+ > \alpha E_{\theta_0}T$. The argument is similar to the proof of Proposition 12.3. From the form of φ_+ ,

$$\begin{aligned} E_{\theta_0}T\varphi_+ - \alpha E_{\theta_0}T &= E_{\theta_0}(T - c_+)\varphi_+ - E_{\theta_0}(T - c_+)\alpha \\ &= E_{\theta_0}(T - c_+)(\varphi_+ - \alpha) \\ &= E_{\theta_0}|T - c_+||\varphi_+ - \alpha|. \end{aligned}$$

Since $\alpha \in (0, 1)$, this expression is strictly positive unless $P_{\theta_0}(T = c_+) = 1$. But if $P_{\theta_0}(T = c_+) = 1$, then $P_{\theta}(T = c_+) = 1$ for all $\theta \in \Omega$ and all distributions in the family are the same. Thus $E_{\theta_0}T\varphi_+ > \alpha E_{\theta_0}T$. Using this in the formula in Theorem 12.20,

$$\beta'_{\varphi_+}(\theta_0) = \eta'(\theta_0)E_{\theta_0}T\varphi_+ - \alpha B'(\theta_0) > \alpha(\eta'(\theta_0)E_{\theta_0}T - B'(\theta_0)).$$

The lower bound here is zero because $B'(\theta) = \eta'(\theta)E_{\theta}T$, which follows from Theorem 12.20, with φ identically one. $\square$

Remark 12.28. Because $B'(\theta) = \eta'(\theta)E_{\theta}T$, using Theorem 12.20 the constraint $\beta'_{\varphi^*}(\theta_0) = 0$ in Theorem 12.26 becomes

$$0 = \eta'(\theta_0)[E_{\theta_0}T\varphi^* - \alpha E_{\theta_0}T] = \eta'(\theta_0)\text{Cov}_{\theta_0}(\varphi^*, T).$$

So any two-sided test φ^* with level α that is uncorrelated with T is uniformly most powerful unbiased.

Example 12.29. Suppose X has an exponential distribution with failure rate θ , so

$$p_\theta(x) = \begin{cases} \theta e^{-\theta x}, & x > 0; \\ 0, & \text{otherwise,} \end{cases}$$

and consider testing $H_0 : \theta = 1$ versus $H_1 : \theta \neq 1$. Let

$$\varphi = \begin{cases} 0, & X \in (c_1, c_2); \\ 1, & X \leq c_1 \text{ or } X \geq c_2. \end{cases}$$

By Theorem 12.26, φ is uniformly most powerful unbiased provided

$$E_1 \varphi = 1 - \int_{c_1}^{c_2} e^{-x} dx = 1 - e^{-c_1} + e^{-c_2} = \alpha \quad (12.8)$$

and

$$\begin{aligned} E_1 X \varphi &= E_1 X - E_1 X(1 - \varphi) = 1 - \int_{c_1}^{c_2} x e^{-x} dx \\ &= 1 - (1 + c_1)e^{-c_1} + (1 + c_2)e^{-c_2} = \alpha E_1 X = \alpha. \end{aligned}$$

Using (12.8), this equation simplifies to

$$c_1 e^{-c_1} = c_2 e^{-c_2}. \quad (12.9)$$

Isolating c_2 in (12.8),

$$c_2 = -\log(e^{-c_1} - 1 + \alpha).$$

Using this in (12.9),

$$c_1 e^{-c_1} = -(e^{-c_1} - 1 + \alpha) \log(e^{-c_1} - 1 + \alpha).$$

The solution to this equation must be found numerically. Note that as c_1 varies from 0 to $-\log(1 - \alpha) > 0$, the left-hand side increases from 0 to $-(1 - \alpha) \log(1 - \alpha) > 0$, as the right-hand side decreases from $-\alpha \log \alpha > 0$ to 0, and so, by continuity, a solution must exist. For $\alpha = 5\%$, $c_1 = 0.042363$ and $c_2 = 4.7652$.

In practice, numerical issues can be eliminated by choosing c_1 and c_2 so that $P_1(X \leq c_1) = P_1(X \geq c_2) = \alpha/2$, for then $c_1 = -\log(1 - \alpha/2)$ and $c_2 = -\log(\alpha/2)$. But the resulting test is biased. For instance, if $\alpha = 5\%$, $c_1 = 0.025318$ and $c_2 = 3.6889$, quite different from the critical values above for the best unbiased test.

12.8 Problems⁶

- *1. Suppose $X \sim P_\theta$ for some $\theta \in \Omega$, and that U is uniformly distributed on $(0, 1)$ and is independent of X . Let $\varphi(X)$ be a randomized test based on X . Find a nonrandomized test based on X and U , so $\psi(X, U) = 1_S(X, U)$ for some critical region S , with the same power function as φ , $E_\theta \varphi(X) = E_\theta \psi(X, U)$, for all $\theta \in \Omega$.
- *2. Suppose $\sup |h(x)| = M$ and $Eh(Z) = 0$, where $Z \sim N(0, 1)$. Give a sharp upper bound for $Eh(2Z)$.
- *3. Determine the density of Z_1/Z_2 when Z_1 and Z_2 are independent standard normal random variables. (This should be useful in the next problem.)
- *4. Let X_1 and X_2 be independent, and let $\sigma_1^2 > 0$ and $\sigma_2^2 > 0$ be known variances. Find the error rate for the best symmetric test of $H_0 : X_1 \sim N(0, \sigma_1^2), X_2 \sim N(0, \sigma_2^2)$ versus $H_1 : X_1 \sim N(0, \sigma_2^2), X_2 \sim N(0, \sigma_1^2)$. A symmetric test here is a test that takes the opposite action if the two data values are switched, so $\varphi(x_1, x_2) = 1 - \varphi(x_2, x_1)$. For a symmetric test the error probabilities under H_0 and H_1 will be equal.
- 5. Suppose $\sup_{x \geq 0} |h(x)| = M$ and $Eh(X) = 0$, where X has a standard exponential distribution. Give a sharp upper bound for $Eh(2X)$.
- *6. Suppose X is uniformly distributed on $(0, 2)$.
 - a) Suppose $\sup_{(0,2)} |h(x)| \leq M$ and $Eh(X) = 0$. Give an upper bound for $Eh(X^2/2)$, and a function h that achieves the bound.
 - b) Suppose instead that $|h(x)| \leq Mx$, $0 < x < 2$, but we still have $Eh(X) = 0$. Now what is the best upper bound for $Eh(X^2/2)$? What function achieves the bound?
- *7. Consider a model in which X has density

$$p_\theta(x) = \frac{\theta}{(1 + \theta x)^2}, \quad x > 0.$$

- a) Show that the derivative of the power function β of a test φ is given by

$$\beta'(\theta) = E_\theta \left[\frac{1 - \theta X}{\theta(1 + \theta X)} \varphi(X) \right].$$

- b) Among all tests with $\beta(1) = \alpha$, which one maximizes $\beta'(1)$?
- 8. Let X have a Poisson distribution with mean one. Suppose $|h(x)| \leq 1$, $x = 0, 1, 2, \dots$, and $Eh(X) = 0$. Find the largest possible value for $Eh(2X)$, and the function h that achieves the maximum.
- *9. Suppose data X has density p_θ , $\theta \in \Omega \subset \mathbb{R}$, and that these densities are regular enough that the derivative of the power function of any test function φ can be evaluated differentiating under the integral sign,

$$\beta'_\varphi(\theta) = \int \varphi(x) \frac{\partial p_\theta(x)}{\partial \theta} d\mu(x).$$

⁶ Solutions to the starred problems are given at the back of the book.

A test φ^* is called *locally most powerful* testing $H_0 : \theta = \theta_0$ versus $H_1 : \theta > \theta_0$ if it maximizes $\beta'_\varphi(\theta_0)$ among all tests φ with level α . Determine the form of the locally most powerful test.

- *10. Suppose $X = (X_1, \dots, X_n)$ with the X_i i.i.d. with common density f_θ . The locally most powerful test of $H_0 : \theta = \theta_0$ versus $H_1 : \theta > \theta_0$ from Problem 12.9 should reject H_0 if an appropriate statistic T exceeds a critical value c . Use the central limit theorem to describe how the critical level c can be chosen when n is large to achieve a level approximately α . The answer should involve Fisher information at $\theta = \theta_0$.
11. Laplace's law of succession gives a distribution for Bernoulli variables $X_1, X_2, \dots$ in which $P(X_1 = 1) = 1/2$, and

$$P(X_{j+1} = 1 | X_1 = x_1, \dots, X_j = x_j) = \frac{1 + x_1 + \dots + x_j}{j + 2}, \quad j \geq 1.$$

Consider testing the hypothesis H_1 that $X_1, \dots, X_n$ have this distribution against the null hypothesis H_0 that the variables are i.i.d. with $P(X_i = 1) = 1/2$. If $n = 10$, find the best test with size $\alpha = 5\%$. What is the power of this test?

12. An entrepreneur would like to sell a fixed amount M of some product through online auctions. Let $R(t) \geq 0$ denote his selling rate at time t . Assuming all of the merchandise is sold eventually,

$$\int_0^\infty R(t) dt = M.$$

The sales rate and price should be related, with the sales rate increasing as the price decreases. Assume that price is inversely proportional to $\sqrt{R}$, so that the rate of return (price times selling rate) at time t is $c\sqrt{R(t)}$. Discounting future profits, the entrepreneur would like to maximize

$$\int_0^\infty c\sqrt{R(t)}e^{-\delta t} dt,$$

where $\delta > 0$ denotes the discount rate. Use a Lagrange multiplier approach to find the best rate function R for the entrepreneur.

13. Consider simple versus simple testing from a Bayesian perspective. Let Θ have a Bernoulli distribution with $P(\Theta = 1) = p$ and $P(\Theta = 0) = 1 - p$. Given $\Theta = 0$, X will have density p_0 , and given $\Theta = 1$, X will have density p_1 .
- a) Show that the chance of accepting the wrong hypothesis in the Bayesian model using a test function φ is

$$R(\varphi) = E[I\{\Theta = 0\}\varphi(X) + I\{\Theta = 1\}(1 - \varphi(X))].$$

- b) Use smoothing to relate $R(\varphi)$ to $E_0\varphi = E[\varphi(X) | \Theta = 0]$ and $E_1\varphi = E[\varphi(X) | \Theta = 1]$.

- c) Find the test function φ^* minimizing $R(\varphi)$. Show that φ^* is a likelihood ratio test, identifying the critical value k .
- *14. Let X denote the number of tails before the first heads if a coin is tossed repeatedly. If successive tosses are independent and p is the chance of heads, determine the uniformly most powerful test of $H_0 : p = 1/2$ versus $H_1 : p < 1/2$ with level $\alpha = 5\%$. What is the power of this test if p is 40%?
- *15. Suppose X and Y are jointly distributed from a bivariate normal distribution with correlation ρ , means $EX = EY = 0$, and $\text{Var}(X) = \text{Var}(Y) = 1/(1 - \rho^2)$. Determine the uniformly most powerful test of $H_0 : \rho \leq 0$ versus $H_1 : \rho > 0$ based on (X, Y) .
- *16. Consider a location family with densities $p_\theta(x) = g(x - \theta)$. Show that if g is twice differentiable and $d^2 \log g(x)/dx^2 \leq 0$ for all x , then the densities have monotone likelihood ratios in x . Give an analogous differential condition sufficient to ensure that densities for a scale family $p_\theta(x) = g(x/\theta)/\theta$, $x > 0$, have monotone likelihood ratios in x .
- *17. *p-values*. Suppose we have a family of tests φ_α , $\alpha \in (0, 1)$ indexed by level (so φ_α has level α), and that these tests are “nested” in the sense that $\varphi_\alpha(x)$ is nondecreasing as a function of α . We can then define the “p-value” or “attained significance” for observed data x as $\inf\{\alpha : \varphi_\alpha(x) = 1\}$, thought of as the smallest value for α where test φ_α rejects H_0 . Suppose we are testing $H_0 : \theta \leq \theta_0$ versus $H_1 : \theta > \theta_0$ and that the densities for data X have monotone likelihood ratios in T . Further suppose T has a continuous distribution.
- Show that the family of uniformly most powerful tests are nested in the sense described.
 - Show that if $X = x$ is observed, the p-value $P(x)$ is

$$P_{\theta_0}[T(X) > t],$$

where $t = T(x)$ is the observed value of T .

- Determine the distribution of the p-value $P(X)$ when $\theta = \theta_0$.
18. Let F be a cumulative distribution function that is continuous and strictly increasing on $[0, \infty)$ with $F(0) = 0$, and let q_α denote the upper α th quantile for F , so $F(q_\alpha) = 1 - \alpha$. Suppose we have a single observation X with
- $$P_\theta(X \leq x) = F(x/\theta), \quad x \in \mathbb{R}, \quad \theta > 0.$$
- Consider testing $H_0 : \theta \leq \theta_0$ versus $H_1 : \theta > \theta_0$. Find the significance level for the test $\varphi = 1_{(c, \infty)}$. What choice for c will give a specified level α ?
 - Let φ_α denote the test with level α in part (a). Show that the tests φ_α , $\alpha \in (0, 1)$, are nested in the sense described in Problem 12.17, and give a formula to compute the p-value $P(X)$.
19. Suppose X has a Poisson distribution with parameter λ . Determine the uniformly most powerful test of $H_0 : \lambda \leq 1$ versus $H_1 : \lambda > 1$ with level $\alpha = 5\%$.

- *20. Do the densities $p_\theta(x) = (1 + \theta x)/2$, $x \in (-1, 1)$, $\theta \in [-1, 1]$, have monotone likelihood ratios in $T(x) = x$?
- *21. Let f be a specified probability density on $(0, 1)$ and let

$$p_\theta(x) = \theta + (1 - \theta)f(x), \quad x \in (0, 1),$$

where $\theta \in [0, 1]$ is an unknown parameter. Show that these densities have monotone likelihood ratios, identifying the statistic $T(x)$.

22. Suppose we observe a single observation X from $N(\theta, \theta^2)$.
- Do the densities for X have monotone likelihood ratios?
 - Let ϕ^* be the best level α test of $H_0 : \theta = 1$ versus $H_1 : \theta = 2$. Is ϕ^* also the best level α test of $H_0 : \theta = 1$ versus $H_1 : \theta = 4$?
23. Consider tests for $H_0 : \theta = 0$ versus $H_1 : \theta \neq 0$ based on a single observation X from $N(\theta, 1)$. Using the apparent symmetry of this testing problem, it seems natural to base a test on $Y = |X|$.
- Find densities q_θ for Y and show that the distribution for Y depends only on $|\theta|$.
 - Show that the densities q_θ , $\theta \geq 0$, have monotone likelihood ratios.
 - Find the uniformly most powerful level α test of H_0 versus H_1 based on Y .
 - The uniformly most powerful test $\varphi^*(Y)$ in part (c) is not most powerful compared with tests based on X . Find a level α test $\varphi(X)$ with better power at $\theta = -1$,

$$E_{-1}\varphi(X) > E_{-1}\varphi^*(Y).$$

What is the difference in power at $\theta = -1$ if $\alpha = 5\%$?

24. Let P_0 and P_1 be two probability distributions, and for $\epsilon \in (0, 1)$, let P_ϵ denote the mixture distribution $(1 - \epsilon)P_0 + \epsilon P_1$. Let E_0 , E_1 , and E_ϵ denote expectation when $X \sim P_0$, $X \sim P_1$, and $X \sim P_\epsilon$, respectively.
- Let φ be a test function with $\alpha = E_0\varphi(X)$ and $\beta = E_1\varphi(X)$. Express $E_\epsilon\varphi(X)$ as a function of ϵ , α , and β .
 - Using the result in part (a), argue directly that if φ is the most powerful level α test of $H_0 : X \sim P_0$ versus $H_1 : X \sim P_1$, then it is also the most powerful level α test of $H_0 : X \sim P_0$ versus $H_1 : X \sim P_\epsilon$.
 - Suppose P_0 and P_1 have densities p_0 and p_1 , respectively, with respect to a measure μ . Find the density for P_ϵ .
 - Using part (c), show that if φ is a likelihood ratio test of $H_0 : X \sim P_0$ versus $H_1 : X \sim P_1$, then it is also a likelihood ratio test of $H_0 : X \sim P_0$ versus $H_1 : X \sim P_\epsilon$.
- *25. Suppose X has a Poisson distribution with mean λ , and that U is uniformly distributed on $(0, 1)$ and is independent of X .
- Show that the joint densities of X and U have monotone likelihood ratios in $T = X + U$.
 - Describe how to construct level α uniformly most powerful tests of $H_0 : \lambda = \lambda_0$ versus $H_1 : \lambda > \lambda_0$ based on X and U . Specify the resulting test explicitly if $\alpha = 5\%$ and $\lambda_0 = 2$.

- c) Describe confidence intervals dual to the family of tests in part (b).
Give the confidence interval if the data are $X = 2$ and $U = 0.7$.
26. Suppose X has a geometric distribution with success probability θ , so $P_\theta(X = x) = \theta(1 - \theta)^x$, $x = 0, 1, \dots$; and that U is uniformly distributed on $(0, 1)$ and is independent of X .
- a) Show that the joint densities of X and U have monotone likelihood ratios in $T = -(X + U)$.
- b) Describe how to construct level α uniformly most powerful tests of $H_0 : \theta = \theta_0$ versus $H_1 : \theta > \theta_0$ based on X and U . Specify the resulting test explicitly if $\alpha = 5\%$ and $\theta_0 = 1/20$.
- c) Describe confidence intervals dual to the family of tests in part (b).
Give the confidence interval if the data are $X = 2$ and $U = 0.7$.
27. Suppose $X_1, \dots, X_n$ are i.i.d. from $N(0, \sigma^2)$.
- a) Determine the uniformly most powerful test of $H_0 : \sigma = \sigma_0$ versus $H_1 : \sigma > \sigma_0$.
- b) Find a confidence interval for σ using duality from the tests in part (a).
- *28. Let $X_1, \dots, X_n$ be i.i.d. observations from a uniform distribution on the interval $(0, \theta)$. Find confidence intervals S_1 dual to the family of uniformly most powerful tests of $\theta = \theta_0$ versus $\theta > \theta_0$ and S_2 dual to the family of uniformly most powerful tests of $\theta = \theta_0$ versus $\theta < \theta_0$. Then use the result from the Problem 9.12 to find a 95% confidence interval for θ . This interval should have finite length and exclude zero.
29. Suppose Y_1 and Y_2 are independent variables, both uniformly distributed on $(0, \theta)$, but our observation is $X = Y_1 + Y_2$.
- a) Show that the densities for X have monotone likelihood ratios.
- b) Find the UMP level α test of $H_0 : \theta = \theta_0$ versus $H_1 : \theta > \theta_0$ based on X .
- c) Find a confidence set for θ dual to the tests in part (b).
30. Let X and Y be independent with $X \sim N(\mu_x, 1)$ and $Y \sim N(\mu_y, 1)$. Take $\|\mu\|^2 = \mu_x^2 + \mu_y^2$, and consider testing $H_0 : \mu_x = \mu_y = 0$ versus $H_1 : \|\mu\| > 0$. For rotational symmetry, a test based on $T = X^2 + Y^2$ may seem natural. The density of T is

$$f_{\|\mu\|}(t) = \begin{cases} \frac{1}{2} I_0(\sqrt{t} \|\mu\|) \exp\left\{-\frac{1}{2}(t + \|\mu\|^2)\right\}, & t > 0; \\ 0, & \text{otherwise,} \end{cases}$$

where I_0 is a modified Bessel function given by

$$I_0(x) = \frac{1}{\pi} \int_0^\pi e^{x \cos \omega} d\omega.$$

- a) Show that $I_0(x) > I'_0(x)$ and that $xI''_0(x) + I'_0(x) = xI_0(x)$.
- b) Show that $xI'_0(x)/I_0(x)$ is increasing in x . Use this to show that for $c > 1$, $I_0(cx)/I_0(x)$ is an increasing function of x . Hint:

$$\log I_0(cx) - \log I_0(x) = \int_1^c \frac{\partial \log I_0(ux)}{\partial u} du.$$

- c) Show that the densities $f_{\|\mu\|}$ have monotone likelihood ratios.
 - d) Find the uniformly most powerful level α test of H_0 versus H_1 based on T .
 - e) Find a level α test of H_0 versus H_1 based on X and Y that has power as high as possible if $\mu_x = \mu_y = 1$. Is this the same test as the test in part (d)?
 - f) Suggest a level α test of $\tilde{H}_0 : \mu_x = c_x, \mu_y = c_y$, versus $\tilde{H}_1 : \mu_x \neq c_x$ or $\mu_y \neq c_y$, based on $\tilde{T} = (X - c_x)^2 + (Y - c_y)^2$.
 - g) Find a $1 - \alpha$ confidence region for (μ_x, μ_y) dual to the family of tests in part (f). What is the shape of your confidence region?
- *31. Suppose $X \sim N(\theta, 1)$, and let φ be a test function with power $\beta(\theta) = E_\theta \varphi(X)$.
- a) Show that $\beta'(0) = E_0 X \varphi(X)$.
 - b) What test function φ maximizes $\beta(1)$ subject to constraints $\beta(0) = \alpha$ and $\beta'(0) = 0$?
- *32. Suppose X_1 and X_2 are independent positive random variables with common Lebesgue density $p_\theta(x) = \theta/(1 + \theta x)^2$, $x > 0$.
- a) Use dominated convergence to write the derivative $\beta'(\theta)$ of the power function for a test φ as an expectation.
 - b) Determine the locally most powerful test φ of $H_0 : \theta \leq \theta_0$ versus $H_1 : \theta > \theta_0$ with $\beta_\varphi(\theta_0) = 5\%$. As in Problem 12.9, a locally most powerful test here would maximize $\beta'(\theta_0)$ among all tests with level α . Hint: The relevant test statistic can be written as the sum of two independent variables. First find the P_{θ_0} marginal distribution of these variables.
 - c) Determine a 95% confidence region for θ by duality, inverting the family of tests in part (b).
- *33. Suppose we have a single observation from an exponential distribution with failure rate λ , and consider testing $H_0 : \lambda = 2$ versus $H_1 : \lambda \neq 2$. Find a test φ^* with minimal level α among all tests φ with 50% power at $\lambda = 1$ and $\lambda = 3$, $E_1 \varphi = E_3 \varphi = 1/2$.
34. Suppose X has a uniform distribution on $(0, 1)$. Find the test function φ that maximizes $E\varphi(X)$ subject to constraints

$$E\varphi(X^2) = E\varphi(1 - X^2) = 1/2.$$

35. Define $\varphi_1 = 1_{[0,1]}$, $\varphi_2 = 1_{[0,1/2]}$, $\varphi_3 = 1_{[1/2,1]}$, $\varphi_4 = 1_{[0,1/3]}$, $\varphi_5 = 1_{[1/3,2/3]}$, $\varphi_6 = 1_{[2/3,1]}$, etc.
- a) If $x \in [0, 1]$, what is $\limsup_{n \rightarrow \infty} \varphi_n(x)$?
 - b) What is $\lim_{n \rightarrow \infty} \int_0^1 \varphi(x) dx$?
 - c) Suppose f is bounded and nonnegative. Find

$$\lim_{n \rightarrow \infty} \int_0^1 \varphi(x) f(x) dx.$$

- d) Suppose $f \geq 0$ and $\int_0^1 f(x) dx < \infty$. Use dominated convergence to show that

$$\lim_{k \rightarrow \infty} \int_0^1 (f(x) - k)^+ dx = 0.$$

- e) Suppose $f \geq 0$ and $\int_0^1 f(x) dx < \infty$. Show that

$$\lim_{n \rightarrow \infty} \int_0^1 f(x) \varphi_n(x) dx = 0.$$

Hint: Note that for any k ,

$$f(x) = \min\{f(x), k\} + (f(x) - k)^+.$$

Use this to find an upper bound for

$$\limsup_{n \rightarrow \infty} \int_0^1 f(x) \varphi_n(x) dx.$$

- f) Let φ be the “zero” test function, $\varphi(x) = 0$, for all x . Do the functions φ_n converge pointwise to φ ?

36. For $n \geq 1$ and $x \in (0, 1)$, define

$$\phi_n(x) = I([\lfloor 2^n x^2 \rfloor \in \{0, 2, 4, \dots\}),$$

and let μ be Lebesgue measure on $(0, 1)$. Find the weak limit of these functions, that is, a function ϕ on $(0, 1)$ such that $\phi_n \xrightarrow{w} \phi$.

- *37. Let φ_n be a sequence of test functions converging pointwise to φ , $\varphi_n(x) \rightarrow \varphi(x)$ for all x .

a) Does it follow that $\varphi_n^2 \xrightarrow{w} \varphi^2$? Prove or give a counterexample.

b) Does it follow that $1/\varphi_n \xrightarrow{w} 1/\varphi$? Prove or give a counterexample.

38. Let X have a Cauchy distribution with location θ , so

$$p_\theta(x) = \frac{1}{\pi[1 + (x - \theta)^2]}, \quad x \in \mathbb{R},$$

and consider testing $H_0 : \theta = 0$ versus $H_1 : \theta \neq 0$. Find a test φ with level $\alpha = 5\%$ that maximizes $E_1 \varphi$ subject to the constraint $E_1 \varphi = E_{-1} \varphi$. Is this test uniformly most powerful unbiased?

39. Suppose X has an exponential distribution with failure rate λ , so $p_\lambda(x) = \lambda e^{-\lambda x}$ for $x > 0$. Determine the most powerful test of $H_0 : \lambda = 1$ or $\lambda = 4$ versus $H_1 : \lambda = 2$ with level $\alpha = 5\%$. The test you derive is in fact the uniformly most powerful test of $H_0 : \lambda \leq 1$ or $\lambda \geq 4$ versus $H_1 : \lambda \in (1, 4)$ with level $\alpha = 5\%$.

40. *Locally most powerful tests in two-sided situations.* Suppose we have a single observation X with density

$$p_{\theta}(x) = \begin{cases} \frac{\theta}{(1 + \theta x)^2}, & x > 0; \\ 0, & \text{otherwise,} \end{cases}$$

where $\theta > 0$. Find a test φ^* of $H_0 : \theta = 1$ versus $H_1 : \theta \neq 1$ with level $\alpha = 5\%$ that maximizes $\beta_{\varphi}''(1)$, subject to the constraint $\beta_{\varphi}'(1) = 0$.

- *41. Suppose X has a binomial distribution with two trials and success probability p . Determine the uniformly most powerful unbiased test of $H_0 : p = 2/3$ versus $H_1 : p \neq 2/3$. Assume $\alpha < 4/9$.
- *42. Let $X_1 \dots X_4$ be i.i.d. from $N(0, \sigma^2)$. Determine the uniformly most powerful unbiased test of $H_0 : \sigma = 1$ versus $H_1 : \sigma \neq 1$ with size $\alpha = 5\%$.

43. Suppose we observe a single observation X with density

$$f_{\theta}(x) = c(\theta)|x|e^{-(x-\theta)^2/2}, \quad x \in \mathbb{R}.$$

- a) Give a formula for $c(\theta)$ in terms of the cumulative distribution function Φ for the standard normal distribution.
- b) Find the uniformly most powerful unbiased test of $H_0 : \theta = 0$ versus $H_1 : \theta \neq 0$.
- *44. Consider testing $H_0 : \theta = \theta_0$ versus $H_1 : \theta \neq \theta_0$ based on a single observation X with density

$$p_{\theta}(x) = \begin{cases} \frac{\theta e^{\theta x}}{2 \sinh \theta}, & |x| < 1; \\ 0, & |x| \geq 1. \end{cases}$$

When $\theta = 0$, p_{θ} should be $1/2$ if $|x| < 1$, and zero otherwise.

- a) Specify the form of the uniformly most powerful unbiased test with level α , and give equations to determine constants needed to specify the test.
- b) Specify the uniformly most powerful unbiased test explicitly when $\theta_0 = 0$.
45. Let $X_1, \dots, X_n$ be independent with

$$X_i \sim N(t_i \theta, 1), \quad i = 1, \dots, n,$$

where $t_1, \dots, t_n$ are known constants and θ is an unknown parameter.

- a) Determine the uniformly most powerful unbiased test of $H_0 : \theta = \theta_0$ versus $H_1 : \theta \neq \theta_0$.
- b) Find a confidence region for θ inverting the family of tests in part (a).
46. Suppose our data consist of two independent observations, X and Y , from a Poisson distribution with mean λ . Find the uniformly most powerful unbiased test of $H_0 : \lambda = 1$ versus $H_1 : \lambda \neq 1$ with level $\alpha = 10\%$.
47. A random angle X has density

$$p_{\theta}(x) = \begin{cases} \frac{\exp[\theta \cos x]}{2\pi I_0(\theta)}, & x \in [0, 2\pi); \\ 0, & \text{otherwise,} \end{cases}$$

where $\theta \in \mathbb{R}$ and I_0 is a modified Bessel function ($I_0(0) = 1$). Find the uniformly most powerful unbiased test of $H_0 : \theta = 0$ versus $H_1 : \theta \neq 0$ with level α .

48. Suppose X has density

$$p_{\theta}(x) = \frac{x^2 \exp[-\frac{1}{2}(x - \theta)^2]}{\sqrt{2\pi}(1 + \theta^2)}, \quad x \in \mathbb{R}.$$

Find the uniformly most powerful unbiased test of $H_0 : \theta = 0$ versus $H_1 : \theta \neq 0$ with level $\alpha = 5\%$.

49. Because a good test of $H_0 : \theta \in \Omega_0$ versus $H_1 : \theta \in \Omega_1$ should have high power on Ω_1 and small power on Ω_0 , a test function ϕ might be chosen to minimize

$$\int_{\Omega_0} \beta_{\phi}(\theta) w(\theta) d\Lambda(\theta) + \int_{\Omega_1} (1 - \beta_{\phi}(\theta)) w(\theta) d\Lambda(\theta),$$

where Λ is a measure on $\Omega = \Omega_0 \cup \Omega_1$ and $w \geq 0$ is a weight function. (With a natural loss structure, Bayes risks would have this form.)

- a) Describe a test function ϕ^* that minimizes this criterion. Assume that $\mathcal{P}$ is a dominated family with densities p_{θ} , $\theta \in \Omega$.
- b) Find the optimal test function ϕ^* explicitly if w is identically one, Λ is Lebesgue measure on $(0, \infty)$, P_{θ} is the exponential distribution with failure rate θ , $\Omega_0 = (0, 1]$, and $\Omega_1 = (1, \infty)$.

